Candidate Name	Centre Number	Candidate Number

For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Ordinary Level

MATHEMATICS SYLLABUS A — 4075/1

PAPER 1 — 2 hours 30 minutes

SPECIMEN PAPER

Additional materials:
Mathematical Instruments

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces at the top of each page.
Check that all the pages are in the booklet and ask the invigilator for a replacement if there are duplicate or missing pages.
If working is needed for any question, it must be shown in the space below that question.
Omission of essential working will result in loss of marks.
Decimal answers which are not exact should be given to three significant figures unless stated otherwise.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.
This paper is marked out of **100**.

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Answer **all** questions.

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS MAY BE USED IN THIS PAPER.

1 **(a)** List the prime numbers which are between 10 and 15.

Answer (a) ……..[2]

(b) Identify an integer from the following numbers: 1; $\sqrt{5}$; $2\frac{3}{4}$ and 3,25.

Answer (b) …… [1]

2 **(a)** Round off $3\frac{3}{4}$ to the nearest whole number.

Answer (a) ……. [1]

(b) Round off 292 to the nearest,

(i) ten,

Answer (b)(i)…… [1]

(ii) hundred.

Answer (b)(ii) …….. [1]

3 **(a)** Simplify

	Hours	Minutes	Seconds
	3	45	30
+	2	20	42

Answer (a)…….. [1]

(b) Express

(i) $2\frac{1}{2}$ days in hours.

Answer (b)(i)……. [1]

(ii) the time 1928 in the 12 hour notation.

Answer (b)(ii)…… [1]

4 A is a set of whole numbers from 2 to 8.
B is a set of prime numbers found between 2 and 8.

(a) Show the above two sets on a well labelled Venn diagram.

Answer (a) …….. [2]

(b) Using set notation, show the relationship between set A and set B.

Answer (b) ……. [1]

5 **(a)** Simplify the ratio 4: 18.

Answer (a)…… [1]

(b) \$88 is shared in the ratio 4: 18.

Find the smaller share.

Answer (b) …… [2]

6 **(a)** Evaluate $\frac{1}{3}$ of 1 200 cm^3,

Answer (a)…… [2]

(a) Simplify $2m^3 - 0{,}5m^3 + 7{,}2m^3$

Answer (b) …… [1]

7 **(a)** Express,

(i) $5{,}9 \times 10^{-2}$ as a number in ordinary form.

Answer (a)(i)…….. [1]

(ii) 80 000 as a number in standard form.

Answer (a)(ii)…… [1]

(b) Find the square root of 49.

Answer (b) …….. [1]

8 Calculate the simple interest on \$180 invested for 2 years at a rate of 5% per annum.

Answer …….. [3]

9 **(a)** Expand 1423_{five} in powers of the given base.

Answer (a)…….. [1]

(b) Convert 43_{ten} to a number in base five.

Answer (b)…….. [1]

(c) Simplify $110_2 + 11_2$, giving the answer in base two.

Answer (c)…… [1]

10 A salesman's monthly salary consists of a basic salary of \$150 and a commission of $2\frac{1}{2}$ % of the total sales of the month.

Calculate his total salary in a month with total sales of \$3 000.

Answer……. [3]

11 Make r the subject of the formula $x = p(q + r)$.

Answer…….. [3]

12 **(a)** Solve the following inequality:

$$2x + 5 < 1$$

Answer (a)……. [2]

(b) Illustrate the answer in **(a)** on a number line.

Answer (b) …… [1]

13 The universal set, ξ, has subsets M and N, such that

$\xi = [1; 2; 3;\ldots;9]$,
$M = [2; 4; 6; 8]$
N is a set of multiples of 3

(a) List all the elements of set

(i) N,

Answer (a)(i)…….. [1]

(ii) $M \cup N$

Answer (a)(ii)……. [1]

(b) Find $n(M \cup N)$

Answer (b)…….. [1]

14 **(a)** **(i)** Simplify –15 + 3

Answer (a)(i)…… [1]

(ii) Evaluate 3 + (–15) ÷ 3

Answer (a)(ii)…….[1]

(b) During the day the temperature was 15°C. At night the temperature fell by 10°C.

Calculate the temperature at night.

Answer (c)…… [1]

15 Solve the following equations:

(a) $3x = 15$

Answer (a)…….. [1]

(b) $6(x + 2) = 19$

Answer (b) …….. [2]

16 Express $\frac{2x-3}{5} + \frac{x+4}{7}$ as a single fraction.

Answer ……[3]

17 Points C and D have coordinates at (4; −1) and (2; 3) respectively.

(a) Find the gradient of the line passing through C and D.

Answer (a)……. [2]

(b) Hence or otherwise find the equation of line passing through C and D.

Answer (b)…….. [1]

18 Solve the simultaneous equations:

$$3x = y + 16$$
$$y - x = -2$$

Answer x = …….

y = ……

[3]

19 **(a)** Given the matrices X = $\begin{pmatrix} 3 & 2 & 1 \\ -1 & 0 & 7 \end{pmatrix}$ and Y = $\begin{pmatrix} 9 & 4 & -6 \\ 8 & -5 & 11 \end{pmatrix}$,

find X + Y.

Answer (a) …….. [2]

(b) Simplify $\begin{pmatrix} -4 & 6 \\ 2 & 5 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix}$

Answer (b) ……. [2]

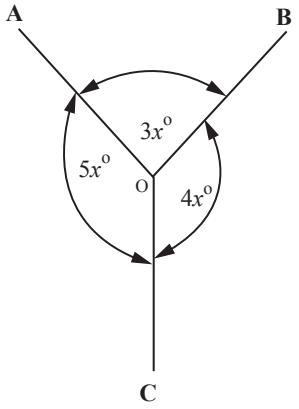
20 **(a)** Express 48 as a product of its prime factors in index form.

Answer (b)……. [1]

(b) Find the Highest Common Factor (HCF) of 14 and 63.

Answer (c)…… [2]

21 The diagram below shows lines AO, BO and CO meeting at a point 0.



(a) Calculate the value of x.

Answer (a)……[2]

(b) Find $A\hat{O}B$

Answer (b)…….. [1]

22 **(a)** Simplify $\frac{2}{3} + \frac{1}{5}$

Answer (a)…….. [2]

(b) Express $\frac{3}{5}$ as a decimal.

Answer (b)……. [1]

(b) Find 45% of 200.

Answer (c)……. [1]

23 A line 2 cm on the map represents an actual distance of 2 km on the ground.

(a) Express the scale in the form 1: n.

Answer (a) …….. [1]

(b) Calculate the length in km on the ground represented by a length of 12 cm on the map.

Answer (b)……. [2]

24 **(a)** Name a triangle with one line of symmetry.

Answer (b)……. [1]

(b) The sum of interior angles of a polygon is 540°.

Find the number of sides of the polygon.

Answer (c)…… [2]

25 Given matrix $\begin{pmatrix} 5 & 5 \\ -1 & 2 \end{pmatrix}$,

(a) Calculate the determinant of the matrix.

Answer (a)…….. [1]

(b) Hence find the inverse of the matrix.

Answer (b)…….. [2]

26 Given the formula $\sqrt{5x - y} = a$,

(a) Make y the subject of the formula.

Answer (a) ……..[2]

(b) Find the value of y when $a = 3$ and $x = 4$.

Answer (b)…… [1]

27 In this question take π to be $\frac{22}{7}$.

A circle has a radius of 7 cm.

Calculate for the circle, the

(a) diameter,

Answer (a)…… [1]

(b) circumference,

Answer (b)…… [2]

(c) area.

Answer (c)…….. [2]

28 The table below shows the frequency table of the distribution of time taken by learners in a class to complete a Mathematical question.

Time (minutes	3	4	5	6	7	8
Frequency	4	7	14	`10	2	2

(a) Calculate the mean time taken by pupils to complete the question.

Answer (a)…… [3]

(b) Pupils who took more than 5 minutes were given another question. A pupil was randomly chosen from the class.

Find the probability that the pupil was given another question.

Answer (b)…….. [2]

29 Matrix P = $\begin{pmatrix} x & 2-x \\ 3 & x-2 \end{pmatrix}$ is singular.

(a) Form an equation in x and show that it reduces to $x^2 + x - 6 = 0$.

Answer (a)……. [2]

(b) Solve the equation $x^2 + x - 6 = 0$.

Answer x = …….. OR

x = ……. [3]

30 On a particular day the official exchange rate of USD (US dollars) to ZWL (ZIG) was 1USD : 28 ZWL.

Convert

(a) 840 ZWL to US Dollar,

Answer (a)……. [2]

(b) 25US Dollar to ZWL (ZIG).

Answer (b)…… [3]